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An embedded level set method for sharp-interface multiphase simulations of Diesel injectors

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ABSTRACT

We propose a comprehensive approach for treating complex wall boundaries in two-phase, free-surface flow simulations on a Cartesian adaptive grid. The external gas–liquid interface is handled by the well-known combined level-set volume-of-fluid (CLSVOF) method. The new element is the coupling with the wall boundary representation using a second level-set function. The no-slip boundary condition at the walls is enforced by properly populating the ghost cells of a narrow band inside the solid body, using a simple and numerically robust treatment of the contact line. In this framework, merging and separation of multiple solid bodies are easily accommodated. Verification tests with grid convergence analysis are presented for a stationary/oscillating body in single-phase flow and for a drop on an inclined plane. Two examples demonstrate the suitability of the proposed approach to study liquid injection. The first is a validation study with data from a scaled-up Diesel injector, to demonstrate how the seamless calculation of internal flow and jet primary atomization can be accomplished. The second is a demonstration of transient atomization response to a measured three-dimensional needle displacement of the injector.

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1. Introduction

The method for time-resolved interface-capturing called CLSVOF (combined level-set volume-of-fluid) was applied, under various implementations, to model bubble and drop dynamics in viscous and viscoelastic environments (Sussman et al., 2007; Stewart et al., 2008; Sussman and Ohta, 2009), ship waves (Sussman and Dommermuth, 2001), and underwater explosions (Kadiouglu and Sussman, 2008). Validation was carried out for sprays formed from jets subject to gas crossflow (Li et al., 2010) or impinging on each other (Arienti et al., 2013), in the latter case with excellent agreement with the statistics derived from experimental measurements.

There are, however, many free-surface flows that also require the management of complex boundary walls. One example is the process of spray formation from liquid injection. The simulation from first principles of spray atomization requires a non-trivial model of the injector in order to correctly define the boundary conditions of the calculation. Under the assumption that internal flow characteristics had limited effects on primary atomization, a simple plug flow velocity profile was assigned as a boundary condition in Arienti et al. (2013). A more realistic inflow, via correlated random velocities with assumed length scale and turbulence intensity, was generated at the orifice exit of a jet injection simulation by Ménard et al. (2007). Still, very few studies have attempted to include the inflow turbulent conditions that result from the actual injector geometry. Notably, in the simulation of jet injection in gas crossflow by Herrmann (2010), the injector was modeled as a short pipe tapering into a flush orifice. A single-phase, pre-computed large eddy simulation of pipe flow was stored as a time sequence of the pre-taper portion of the injector in the subsequent two-phase simulation.

In this paper we develop a general approach within the CLSVOF framework to include the whole injector geometry in a primary atomization simulation. This is accomplished by introducing a second level set function to represent the injector's walls in addition to the level set used to capture the gas–liquid interface. This approach is a valid alternative to boundary-fitted methods, where issues of grid deformation, re-generation and interpolation at each physical timestep can become critical in the case of moving walls. The solution we propose is to let solid boundaries and phase interfaces have unrestricted motion across underlying fixed grid lines. Without the constrain of a conformal mesh, the choice of the computational grid can be optimized for free-surface flow by selecting an isotropic and equispaced (Cartesian) grid.

The most delicate and time consuming operation with Cartesian methods becomes the intersection of the solid body with the regular grid. For single phase flow, several efficient algorithms for

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handling cut cells exist (for instance, in Aftosmis et al., 1997). Since cells cut by the solid wall can be arbitrarily small, explicit update schemes become overly restrictive for time-dependent problems, requiring either to extend the difference stencil of the spatial terms (Berger and LeVeque, 1990), or to use a cell-merging approach (Bayyuk et al., 1993).

With a moving or deforming solid, the second issue for Cartesian methods is how to enforce the correct velocity boundary condition at the interface. In the immersed boundary method originated by Peskin (1972), the flow velocity is obtained from a set of body forces distributed over the nearby field. This approach is well established for single-phase flow, where a high order of accuracy can be achieved. Fadlun et al. (2000) developed a second-order accurate immersed boundary method, on a regular grid, for unsteady three-dimensional flows in complex geometries; velocity interpolation algorithms were developed for prescribing the feedback forcing at the first gridpoint outside the boundary. To determine the velocity and pressure values of the computational cells emerging from solid body motion, Yang and Balaras (2006) introduced a field-extension strategy that recovers a sharp boundary wall instead of a smeared interface. A robust and fast method for cell tagging and shortest distance calculation with an immersed triangulation is discussed by Yang and Stern (2013).

In the development of methods for two-phase flow, an additional layer of complexity emerges because of the numerical treatment of the contact line. At the intersection of the gas–liquid interface with the solid boundary, the contact line must be allowed to move, even if such motion is a mathematical paradox because of the no-slip boundary at the solid surface. This situation in general results in a degradation of the convergence properties of single-phase algorithms for solid boundary treatment.

In the body-conformal finite element method by Baer et al. (2000), the slip is imposed at the mesh nodes forming the contact line. In the two-dimensional sharp-interface approach by Liu et al. (2005), the level set field in the vicinity of the contact line is redistanced in order to impose a specified contact angle; the slip condition is imposed on grid points in the vicinity of the contact line, while a ghost fluid method treatment is used for gas–liquid interfaces. In the level-set only method for sharp interfaces and arbitrary boundaries by Krishnan et al. (2006), a local, two-dimensional level-set field is reconstructed by fitting the interface to a parabolic curve that intersects the solid surface at exactly the given contact angle. The embedded boundary approach by Yang and Balaras (2006) was later complemented by a level-set based ghost-fluid method to treat the gas–liquid interface in the study of wake–ship interaction (Yang and Stern, 2009).

In the volume-of-fluid (VOF) continuum surface force (CSF) method by Bussmann, the contact line slip is achieved implicitly because the advection scheme for the liquid volume fraction utilizes face-centered velocities (Afkhami and Bussmann, 2009); in this way, the center of the cell is removed one half cell width away from the wall. A similar approach is used in this paper, but with the extension that a solid boundary can be arbitrarily positioned with respect to the Cartesian mesh (whereas in Afkhami and Bussmann the solid wall coincides with the boundary domain). In fuel injection simulations, a level-set based ghost-fluid method for the gas–liquid interface with sharp solid wall treatment was explored by Noël et al. (2012) and by Arienti and Sussman (2012).

We propose to augment the CLSVOF method with a second level set function to capture the motion of multiple solid boundaries of arbitrary complexity. The level set framework enables contact, merging, and separation of solid boundaries in a straightforward manner. The use of a level set function to track a moving solid boundary is not new (see, for instance, Arienti et al., 2003), but what we add in this paper is the study of the free surface interaction with it.

The methodology we propose for the treatment of the contact line is somewhat simpler than the one by Afkhami and Bussmann (2009), but it is demonstrated to be robust for an arbitrary position of the solid wall in two and three dimensions. There is also no need for an extension of the pressure field, as required in the methods of Yang and Stern (2009) and Noël et al. (2012). Given these simplifications, the outcome of the verification tests presented in this paper is rather satisfactory, opening the way to future algorithmic improvements. The validation study with an actual Diesel injector, presented later, is also relevant, since there appears to be very few studies on the effects of non-trivial orifice geometry on spray formation.

The numerical aspects of the embedded solid boundary algorithm are described first. The method is then verified with a single-phase crossflow passing over a half cylinder at low Reynolds number; the vorticity field arising from the interaction with the curved wall is compared with the results from more specialized, higher-order methods for single-phase flow, showing the lack of computational artifacts and acceptable rate of convergence. Next, the shape of a drop on a wall surface is calculated for different contact angles. The new element in these tests is that the wall is at an angle with respect to the Cartesian axes. The convergence properties of the final drop shape and the rate of volume conservation are discussed, including dynamic cases where the initial drop shape is different than the final one.

For validation purposes, we consider an early experimental study by Arcoumanis et al. (1998) where the flow velocity inside a scaled-up Bosch six-hole transparent Diesel injector was measured with laser Doppler velocimetry for evaluating average and fluctuating components. The asymmetric spray obtained from the simulation is briefly discussed to point out the relevance of an injector simulation that can capture both internal and external flow.

The last example concerns a transient injection where the injector's needle moves relative to its cap. The geometry and motion data belong to a real injector, reported by Kastengren et al. (forthcoming) and made available by the extensive data set of the Engine Combustion Network (ECN); the calculation demonstrates the ease of the proposed methodology in dealing with moving walls and a changing topology (merging and separation) of the solid boundary.

2. Numerical method

The Navier–Stokes equations for incompressible flow of two immiscible fluids are solved with the one-fluid approach according to the level-set equations for multiphase flow (Chang et al., 1996):

$$\rho(\phi)\frac{D\mathbf{u}}{Dt} = -\nabla p + 2\nabla(\mu(\phi)\mathbf{D}) - \sigma\kappa\nabla H(\phi), \tag{1}$$

$$\nabla\cdot\mathbf{u} = 0, \tag{2}$$

$$H(\phi) = \begin{cases} 1 & \phi \geqslant 0 \\ 0 & \phi < 0, \end{cases} \tag{3}$$

$$\rho(\phi) = \rho_L H(\phi) + \rho_G(1 - H(\phi)), \tag{4}$$

$$\mu(\phi) = \mu_L H(\phi) + \mu_G(1 - H(\phi)), \tag{5}$$

$$\frac{D\phi}{Dt} = 0. \tag{6}$$

In the equations, $\mathbf{u}$ is the vector field, p the pressure, ϕ the level-set function, κ the interface curvature, and $\mathbf{D}$ the deformation tensor, $\mathbf{D} = (\nabla\mathbf{u}+(\nabla\mathbf{u})^T)/2$; H is the Heaviside function and D/Dt the material derivative; σ is the surface tension coefficient. The smooth

zero level of ϕ represents the time-evolving gas–liquid interface. The properties of density, ρ, and dynamic viscosity, μ, are function of ϕ everywhere in the computational domain. The same constant temperature is used for both the liquid and the gas phase.

In the coupled level-set/volume-of-fluid (CLSVOF) method, the advection equation for the liquid volume fraction,

$$\frac{DF}{Dt} = 0, \tag{7}$$

is solved in step with ϕ so that the level-set can be re-initialized in a local mass-preserving fix; the liquid volume fraction F is used with the interface normal from ϕ to construct a volume-preserving distance function.

The nonlinear advection terms in Eq. (1) are calculated in non-conservative, semi-Lagrangian unsplit form. Variables are located according to the staggered MAC grid arrangement; see Fig. 1. A second velocity field for the liquid phase is extrapolated to the gas phase to advect the VOF function (Sussman et al., 2007). The liquid velocity is numerically divergence-free in the liquid region (but not necessarily divergence-free in the gas region, which might cause small over- or under-shoots in the VOF function). A full account of the CLSVOF interface capturing method and of the multiphase flow solver developed by Sussman can be found elsewhere (Sussman et al., 2007), together with verification studies of surface tension-driven oscillations of a spherical drop and of ligament pinch-off induced by capillary instability.

2.1. Solid wall level set

The solid boundary is represented as the zero level set of a second signed distance function, ψ. By convention, the field ψ is positive outside the body and negative inside it. The absolute value of ψ is the minimal distance between the cell centroid and the surface of the body. This function is calculated using the solid boundary description, which may consist of a parameterized curve or of a triangulation listing the surface nodes and their connectivities. In the latter, more general case, the calculation sweeps over all the triangles comprising the body surface and must account for the possibility that the minimal distance occurs either at the corners of a triangle, along one of its edges, or inside it.

The methodology described next works for the solid geometries considered in this paper, but the reader interested in a more efficient approach is referred to the report by Mauch (2000). Here, for simplicity of implementation and to increase execution speed, the ψ calculation is carried out on an auxiliary Cartesian box, labeled S, which is sized to contain the entire solid body. The result is the auxiliary $\psi_{I,J,K}$; indices in capital letters are used to distinguish the function defined on the auxiliary grid from the local $\psi_{i,j,k}$ function defined on a grid block (see Section 2.4 for the description of the block-structured computational environment). The procedure to find the signed distance function from the

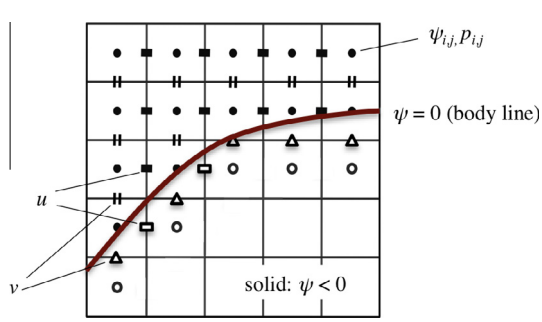

Fig. 1. Schematics for face velocity treatment near a solid wall for the MAC arrangement on a Cartesian grid.

Lagrangian elements, expressed as a list of elements and node coordinates, is as follows:

1. For elements containing more than 3 nodes, break up the element into multiple triangles so that each element has exactly 3 nodes.
2. Traverse all elements: for a given triangle, if one of its sides exceeds the mesh spacing of the auxiliary grid, Δx_S, then split the largest side of that element in half in order to form two new elements. Repeat until no more triangles exist with sides larger than Δx_S.
3. Initialize the signed distance function from the refined element/node list to a large positive value. Also, initialize a tag variable to zero in all the cells.
4. Calculate the wall node normal $\mathbf{n}_{node}$ as the average of the normals $\mathbf{n}_{element}$ of the adjoining triangles to the node. When required, derive the edge normal $\mathbf{n}_{edge}$ from linear interpolation of the node normals from the two nodes that make up the edge.
5. Traverse all the refined Lagrangian elements:
   a. For each triangle, find the Cartesian cell (I, J, K) whose center is closest to the triangle's centroid.
   b. Traverse the $3 \times 3 \times 3$ stencil about cell (I, J, K). For each cell (I', J', K'), check if the projection from its center onto the plane of the triangle is in the interior of the triangle:
   i. If "yes", compute $d = (\mathbf{x}_{I',J',K'} - \mathbf{x}_{centroid}) \cdot \mathbf{n}_{element}$. If tag = 0 or $|d| < |\psi_{I',J',K'}|$, then set $\psi_{I',J',K'} = d$ and tag = 1;
   ii. If "no", then check if the projection is in the interior of the triangle's edge for each of the three edges. If "yes", find the distance from the cell center to its projection on the edge, $d_{project}$. If tag = 0 or $|d_{project}| < |\psi_{I',J',K'}|$ then set $\psi_{I',J',K'} = d_{project} \cdot \mathrm{sign}((\mathbf{x}_{I',J',K'} - \mathbf{x}_{projection}) \cdot \mathbf{n}_{edge})$ and tag = 1;
   iii. If "no", then the closest distance from the cell center to the element must be at one of the element's three nodes. For each node compute the distance d_{node}. If tag = 0 or $|d_{node}| < |\psi_{I',J',K'}|$ then set $\psi_{I',J',K'} = d_{node} \cdot \mathrm{sign}((\mathbf{x}_{I',J',K'} - \mathbf{x}_{node}) \cdot \mathbf{n}_{node})$ and tag = 1.
6. Extend the signed distance function from the cells that have been initialized into the cells that are not:
   a. Set tag = 1 in the cells where the signed distance has already been calculated (a one cell narrow band on either side of the Lagrangian interface) and tag = 0 everywhere else.
   b. Extend the distance in a broader band of half-width $\Delta x \cdot (n_{\max} + 4)$.
   i. For a current zero-tag cell, calculate the distance function from the initialized values (from the previous algorithm iteration) that are found in its "star" stencil (in three dimensions, the 7-point stencil of the cubic cell and the six cells sharing a face with it):

$$\left|\psi^{next}_{I,J,K}\right| = \Delta x + \sum_{\substack{stencil \\ tag = 1}} \left|\psi^{previous}_{I',J',K'}\right| \Big/ \sum_{stencil} tag^{previous}_{I',J',K'}. \tag{8}$$

   ii. Set $tag^{next}_{I,J,K} = 1$.

   c. Repeat until there are no more zero-tag cells in the band. Assign $\psi_{I,J,K} = \Delta x \cdot (n_{\max} + 4)$ everywhere else.
   d. Extend the sign of ψ from the narrow-band cells whose sign was calculated in step 5:
   i. Reset tag to zero outside the narrow band.
   ii. For a current zero-tag cell, calculate from the 7-point stencil

$$sign_sum = \sum_{\substack{stencil \\ tag = 1}} \mathrm{sign}\left(\psi^{previous}_{I',J',K'}\right) \Big/ \sum_{stencil} tag^{previous}_{I',J',K'}. \tag{9}$$

iii. Set $\psi^{next}_{I,J,K} = \text{sign}(sign_sum) \cdot \left|\psi^{next}_{I,J,K}\right|$ and $tag^{next}_{I,J,K} = 1$.
iv. Repeat until there are no more zero-tag cells.

In case of moving boundaries, whether a cell is inside or outside the embedded boundary only needs to be determined in the narrow band: if $|\psi| > 2\Delta x$ at the previous time step, then the sign of the distance function can be expected not to change in the following time step. Thus, the conversion from the Lagrangian to the Eulerian description could be accelerated by reusing the signed distance from the previous time step. As the algorithm is implemented now, however, the signed distance extension is carried out at every time step if the distance function needs to be recalculated.

When N solid bodies exist, the corresponding level set functions are generated in sequence, one per body: ψ^1 and $\psi^2, \ldots, \psi^N$. The functions are then combined on S according to the relation

$$\psi = \min(\psi^1, \psi^2, \ldots, \psi^N). \tag{10}$$

The reason why the minimum is sought in this operation is the convention that $\psi > 0$ in the fluid region. Evaluating Eq. (8) cell by cell makes it particularly easy to merge solid bodies without the issues of connectivity or of skewed computational cells that one could find in a conformal mesh.

The last step is to quadratically interpolate the auxiliary $\psi_{I,J,K}$ onto the local $\psi_{i,j,k}$ of a grid block. In case of moving wall boundaries, $\psi_{I,J,K}$ needs to be calculated at every timestep; for stationary solid bodies, $\psi_{I,J,K}$ is only calculated at the beginning of the simulation, whereas $\psi_{i,j,k}$ is interpolated at run-time as requested by adaptive mesh refinement. The CPU timing of this algorithm is discussed for two cases of fixed and moving solid boundaries in Sections 4 and 5, respectively.

2.2. Boundary conditions at the wall

The components of flow velocity at the wall are made consistent with the no-slip boundary condition through the Poisson equation and the projection operator. The solid wall velocity, $\mathbf{v}$, is first stored for all the Lagrangian nodes. Then, for each cell of the auxiliary grid whose center $\mathbf{x}_{I,J,K}$ is the closest to an element of the tessellation, the value $\mathbf{v}_{I,J,K}$ is assigned as a weighted average of the values of the three nodes belonging to that element; the weights are defined as the normalized inverse distance of $\mathbf{x}_{I,J,K}$ to the nodes positions.

At this point, $\mathbf{v}_{I,J,K}$ is only known in a few cells of the box S. The values of $\mathbf{v}_{I,J,K}$ are then extended from the solid surface by a simple front-advancing procedure where an unmarked cell takes its value from the neighboring cells that have already been marked. The procedure is concluded by the quadratic interpolation of the auxiliary variable $\mathbf{v}_{I,J,K}$ to obtain the local centroid value $\mathbf{v}_{i,j,k}$.

In transferring the velocity field to the cell faces, it is convenient to refer to Fig. 1, which provides a two-dimensional schematics of the conventional MAC grid: cell and face values that need to be populated at every iteration are indicated by empty symbols. For a cell in the fluid region, $\psi_{i,j,k} > 0$, a solid face is a face such that the solid level set function is negative in at least one of the adjoining cell centers: for instance, $\psi_{i,j,k} \cdot \psi_{i+1,j,k} < 0$. The velocity at the $i + 1/2$ face is then calculated as the average of the cell values $\mathbf{v}_{i,j,k}$ and $\mathbf{v}_{i+1,j,k}$.

Note that the boundary condition for the pressure projection step can be formally written as

$$\nabla p \cdot \mathbf{n}_{face} = \rho_{i+1/2,j,k}\left(\mathbf{u}^*_{i+1/2,j,k} - \mathbf{v}_{i+1/2,j,k}\right) \cdot \mathbf{n}_{face} = 0, \tag{11}$$

where $\mathbf{u}^*$ is the provisional field computed from the velocity and volume fraction fields of the previous time step. Thus, in the actual implementation, the Poisson equation is solved under the conditions that $\mathbf{u}^* = \mathbf{v}$ and $\nabla p \cdot \mathbf{n}_{face} = 0$ at solid faces.

2.3. Curvature and contact angle

In calculating curvature at the gas–liquid interface, the "method of heights" based on the level set function ϕ is followed. This procedure can be described for simplicity in two dimensions (see Figs. 1 and 2), but it is implemented in the code also for three dimensions. The use of the level set function in the method of heights was introduced by Sussman and Ohta (2009) in a departure from the previous approach in the CLSVOF method (Sussman et al., 2007; Stewart et al., 2008) based on volume fraction. Sussman and Ohta (2009) suggest that computing the curvature from the level set function is easier in the context of an adaptive hierarchy of grids and slightly more accurate, especially when two interfaces come in close proximity to each other.

Considering two dimensions for simplicity, a 5-point stencil needs to be considered for cell (i, j). Curvature is computed if

$$\phi_{i,j} \cdot \phi_{i',j'} \leqslant 0 \quad \text{and} \quad |\phi_{i,j}| \leqslant |\phi_{i',j'}|, \tag{12}$$

where (i', j') belongs to the stencil. To determine the heights h_{i-1}, h_i, and h_{i+1}, it is then necessary to look at where the sign of ϕ changes, e.g., to find h_i one has to consider the values ϕ_{i,j^*} with $j^* = j - 3, \ldots, j + 3$.

To implement the contact between the gas–liquid interface and the solid wall, the algorithm for curvature reverts to calculating the divergence of the interface normals,

$$\kappa = \nabla \cdot \mathbf{n}_\phi = \nabla \cdot (\nabla\phi/|\nabla\phi|), \tag{13}$$

if any cell of the height function stencil is in the solid domain, $\psi'_{i'}, j' < 0$. The finite difference formulation, albeit less accurate than the method of heights, enables the straightforward enforcing of a prescribed contact angle θ. The 3×3 stencil necessary for this calculation is represented in Fig. 2 by the dashed line. Finite differences between adjacent centers are used to calculate the normal vectors $\mathbf{n}_\phi$ at the corners, identified by crosses in the schematics.

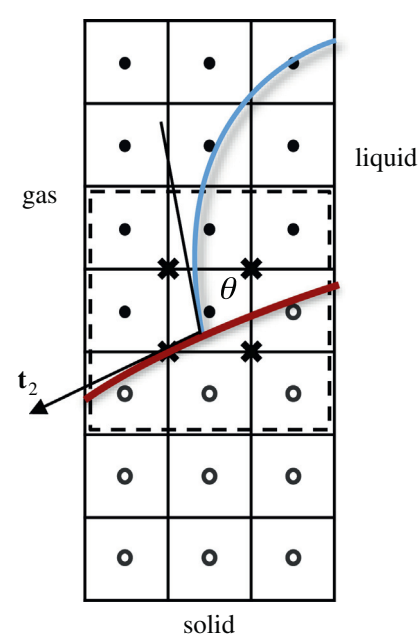

Fig. 2. Contact angle schematics. The zero ψ iso-contour (thick red line) represents the solid wall and the zero ϕ iso-contour (thinner blue curved line) represents the gas–liquid interface. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

If one of the cell centers falls in the solid region, $\mathbf{n}_\phi$ is replaced by the normal determined from θ and from the normal to the local zero solid level set, $\mathbf{n}_\psi = \nabla\psi/|\nabla\psi|$. For this, auxiliary vectors $\mathbf{t}_1$ and $\mathbf{t}_2$ need to be calculated,

$$\mathbf{t}_1 = \mathbf{n}_\psi \times \mathbf{n}_\phi, \tag{14}$$

and

$$\mathbf{t}_2 = \mathbf{n}_\psi \times \mathbf{t}_1. \tag{15}$$

The vector $\mathbf{t}_2$ is tangential to the solid surface and points away from the liquid phase by construction. The contact normal is calculated as

$$\tilde{\mathbf{n}}_{cont} = \mathrm{sign}(\mathbf{n}_\phi \cdot \mathbf{t}_2)\sin\theta\frac{\mathbf{t}_2}{|\mathbf{t}_2|} - \cos\theta\mathbf{n}_\psi, \tag{16}$$

with

$$\mathbf{n}_{cont} = \tilde{\mathbf{n}}_{cont}/|\tilde{\mathbf{n}}_{cont}|. \tag{17}$$

The process is concluded by evaluating the curvature at the face centroids. Surface tension is only applied where (i) ϕ changes sign across adjoining cells and (ii) ψ is positive. Because cell face velocities are used, the slip length is always $\Delta x/2$.

2.4. Adaptive mesh refinement

Another component of the simulation capability presented in this paper is the use of block-structured, adaptive mesh refinement (AMR) for computationally intensive calculations. Starting from the base level, boxes (with a minimum size of, say, 32^3 cells) are combined to cover all the tagged cells within the assigned coverage efficiency. This new set of blocks with the same grid spacing forms level 1. The refinement ratio between two consecutive levels is two. The new level can in turn be tagged for refinement, and the process is repeated until the input grid resolution is achieved. Data on the fine level are either copied from a previous time step or, when the grid structure has changed locally, they are conservatively interpolated from the underlying coarse level. The gas–liquid interface is always embedded in the finest grid level to avoid gross interpolation errors (Kadiouglu and Sussman, 2008).

The definitions and operators necessary to carry out the AMR tasks are provided by the BOXLIB library (CCSE, 2012), which is developed and maintained by the Center for Computational Sciences and Engineering group at Lawrence Berkeley National Laboratories.

The advective terms are calculated on all levels, and the updated data on a fine level are averaged down to the underlying coarser one. The sparse matrix system that results from discretizing the pressure correction equation with discontinuous coefficients on the AMR blocks is solved by the multigrid preconditioned conjugate gradient (MGPCG) method. Compared to our previous studies on the topic of spray atomization (Li et al., 2010; Arienti et al., 2013), the current work takes advantage of a new MGPCG method by Duffy et al. (unpublished), which is specifically developed for the adaptive hierarchy of grids. With an optimal blocking factor, it is found that the new MGPCG AMR method no longer consumes the majority of CPU time per timestep as the number of adaptive levels increases.

3. Verification tests

3.1. Crossflow over a half cylinder

A verification problem for which there is a large pool of numerical and experimental results is the calculation of the two-dimensional flow around a circular cylinder of diameter D. The purpose of this test is to show that the embedded solid boundary technique does not generate artifacts and that the convergence is at least first-order. Flow in the far field is assumed to be uniform, $u = U_\infty$ for $t > 0$. The Reynolds number is defined as $\mathrm{Re} = DU_\infty/\nu$. At sufficiently small values of Re, as in this test, the flow can be considered symmetrical and only a half cylinder is used. The half cylinder is located at $x = 10D$ and $y = 0$ in a $18D \times 6D$ Cartesian domain with symmetry boundary condition along the bottom side.

A medium and a fine grid resolutions are obtained by halving Δx in both coordinate directions, starting from the coarsest grid of 64×192 cells. An even finer grid resolution is calculated with 512×1536 cells.

A close-up of the contours of vorticity around the half-cylinder, both inside and outside the solid boundary, is shown in Fig. 3 at Re = 20. The zero iso-contour of the solid level-set corresponds to the red line. No spurious effects due to the embedded boundary implementation are observable, and the plot qualitatively agrees with the one obtained by Calhoun (2002) and based on solving the streamwise vorticity equations. The dotted, dashed and continuous lines correspond to the coarse (64/6 cells per unity length), intermediate (128/6) and fine grid resolution (256/6). The finest grid resolution (512/6) coincides almost exactly with the fine grid resolution and it is not shown in the plot.

The convergence properties for this case are listed on Table 1. Since there is no closed-form solution for this flow, we evaluate convergence by comparing two simulations at consecutive grid refinements. The norms are $l_2 = \left(\sum_{i=1}^N \delta_i^2\right)^{1/2}$, $l_1 = \sum_{i=1}^N |\delta_i|$ and $l_\infty = \max_N|\delta_i|$, where $\delta = \{\delta_i\}$ is the vector difference of the field values (the two components of velocity and vorticity) taken from two consecutive grid resolutions; N is the total number of grid points of the coarser of the two grids. Thus, the column in Table 1 that is labeled "fine–xfine" lists the norms of the difference vector between the 256×768 grid and the 512×1536 grid, with one grid point every two skipped on the latter. Finally, the last column in Table 1 is the convergence rate calculated from the slope of the linear fit of the three points listed on each row. Note that the l_1 and l_2 norms are divided by N in Table 1. The l_∞ norm shows most clearly that the convergence rate is of first-order for the two components of velocity and of zeroth-order for vorticity. The computational cells with the largest difference between two consecutive grid resolutions are located, as expected, at the boundary of the solid body.

The last element of this test is the evaluation of the reattachment length L from the cylinder's trailing edge. Values at Re = 20

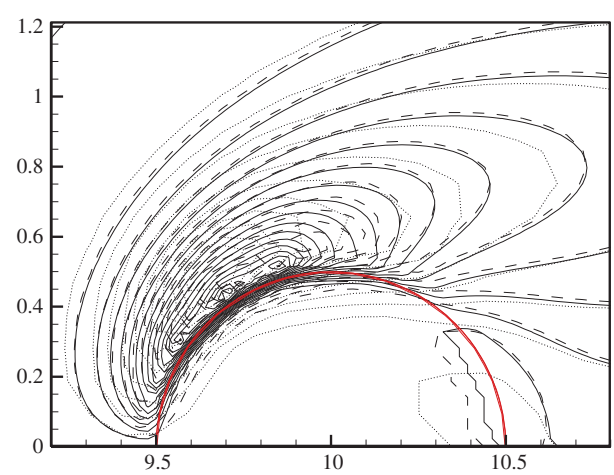

Fig. 3. Contours of vorticity around a half-cylinder in crossflow at Re = 20. The contours, from −6 to 0 in steps of 0.2, are superimposed to the zero solid level-set line (in red), which represents the boundary of the cylinder. The dotted, dashed and continuous lines correspond to the coarse (64/6 cells per unity length), intermediate (128/6) and fine resolution (256/6). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Error norms and convergence rates for the two components of velocity (u_x, u_y) and vorticity ω_z for pairs of increasingly finer grids.

	Coarse–medium	Medium–fine	Fine–xfine	Convergence rate
u_x				
l_2/N	0.0011935	3.2936e−4	2.7023e−5	2.73
l_1/N	0.027651	0.015710	0.0022279	1.82
l_∞	0.14596	0.070753	0.035915	1.01
u_y				
l_2/N	7.5492e−4	8.2722e−5	1.1274e−5	3.03
l_1/N	0.069099	0.0044771	0.0009688	3.08
l_∞	0.14596	0.041539	0.035915	1.15
ω_z				
l_2/N	0.0055727	0.0016805	0.0005150	1.72
l_1/N	0.064975	0.028570	0.0098922	1.36
l_∞	1.81089	2.22074	1.81129	−1.60e−4

and Re = 40 are listed on Table 2 as a function of grid spacing. The differences of consecutive values of L are 0.251, −0.086 and 0.018 for Re = 20; and 0.503, −0.214 and 0.039 for Re = 40. These differences correspond, in both cases, to a more than linear convergence. The numerical results of Calhoun (2002) are also listed on Table 1: the reattachment lengths measured in the experiment by Dennis and Chang (1970) fall between the results from the present calculations and the values reported by Calhoun (2002).

3.2. Oscillating cylinder

The cylinder is now subject to the horizontal oscillation of frequency f

$$x = -A\sin(2\pi f t). \tag{18}$$

The parameters defining this flow are the Keulegan–Carpenter number, $KC = U_{max}/(fD) = 2\pi A/D$ and the Reynolds number, $Re = U_{max}D/\nu$. The values used here, KC = 5 and Re = 100, correspond to a regime of periodic, symmetric vortex shedding.

This test is relevant because later we want to look at multiphase problems with a moving solid boundary. Unphysical, temporal oscillations of the pressure fields have been reported for non boundary-conformal methods when a solid wall moves or deforms. The primary source of this error is that some computational cells belonging to the solid suddenly become active fluid cells. Seo and Mittal (2011) make the case that the violation of local mass conservation at the immersed boundary causes these spurious pressure oscillations.

In Fig. 4, the drag coefficient derived from an oscillating cylinder calculation is compared with results from Dütcsh et al. (1998). The calculation is started from a quiescent field and integrated in time for 3 cycles, until periodic vortex shedding is established. The computational domain is $16D \times 8D$, with $\Delta x = 0.0156D$ and $\Delta t = 0.008\, D/U_{max}$. The cylinder is located at $x = 8D$ and $y = 0$.

The good agreement with the calculation by Dütcsh et al. (1998) is encouraging, considering that their result was obtained with the much more problem-specific configuration of a body-conformal grid oscillating with the cylinder and that the wall grid resolution in the two cases is comparable. The pressure artifact is rather modest for this particular choice of Δx and Δt, even though the drag

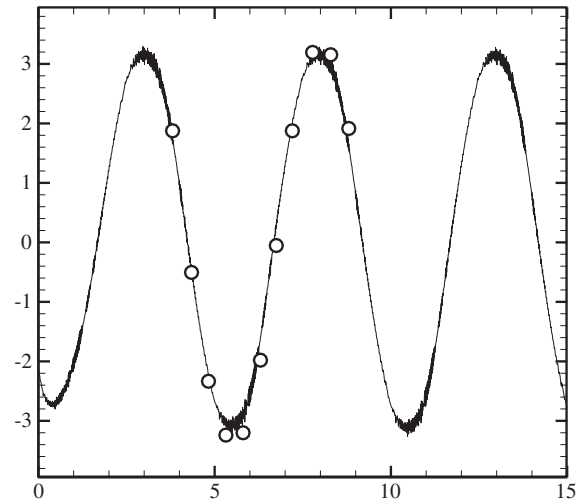

Fig. 4. Drag coefficient as a function of time. Comparison with results from Dütcsh et al. (1998).

coefficient exhibits some ringing, visible at the crests and troughs of the curve of Fig. 4. In agreement with the error analysis by Seo and Mittal (2011), we indeed observed increasing spurious oscillations when the same simulation is run with a much smaller timestep but with the same grid spacing.

The vorticity and pressure flow fields, as shown in Fig. 5, are also devoid of spurious effects. As the cylinder moves to the left, a thin boundary layer develops from the upper wall and eventually detaches in a vortex (the same process would occur symmetrically on the bottom half of the cylinder). Vortex generation stops when the body reaches its extreme left location, as shown in frames (c and d). Then the cylinder moves backwards, and the same process takes place on its right side. The flow fields in Fig. 5 compare well with the analogous plots in Dütcsh et al. (1998) and Yang and Balaras (2006), and indicate that the present method can properly capture the dynamics of the vorticity field.

This and the previous test suggest a convergence rate that is approximately linear with grid refinement. A first-order convergence rate, low if compared to existing methods specialized for single-phase flow, is not a concern as we tackle the interaction of a gas–liquid interface with a solid boundary. In that more challenging context, we will show that linear convergence or higher is maintained, at least in the case where the static contact angle is 90°.

3.3. Static drop on a horizontal plane

We first consider a half-drop on a plane aligned with one coordinate direction. Simulations are carried out first in two dimensions for ease of discussion. As in all the other tests in this section, the density of the liquid is 1000 kg/m^3 and the density of the gas is 1.20 kg/m^3. The viscosities of the two phases are 0.0010 kg/m-s and $1.78 \cdot 10^{-4}$ kg/m-s, respectively. Surface tension is 0.0709 N/m. A symmetry plane is established at the left of the domain, and outflow boundary conditions at the other sides. The computational domain is a square with 32×32 cells. For contact

Table 2
Reattachment length L evaluated from the cylinder's trailing edge as a function of grid spacing.

	$\Delta x/D$=				Dennis and Chang (1970)	Calhoun (2002)
	0.09375	0.04688	0.02344	0.01172		
L (Re = 20)	0.825	1.076	0.990	1.008	0.94	0.91
L (Re = 40)	2.130	2.633	2.419	2.458	2.35	2.18

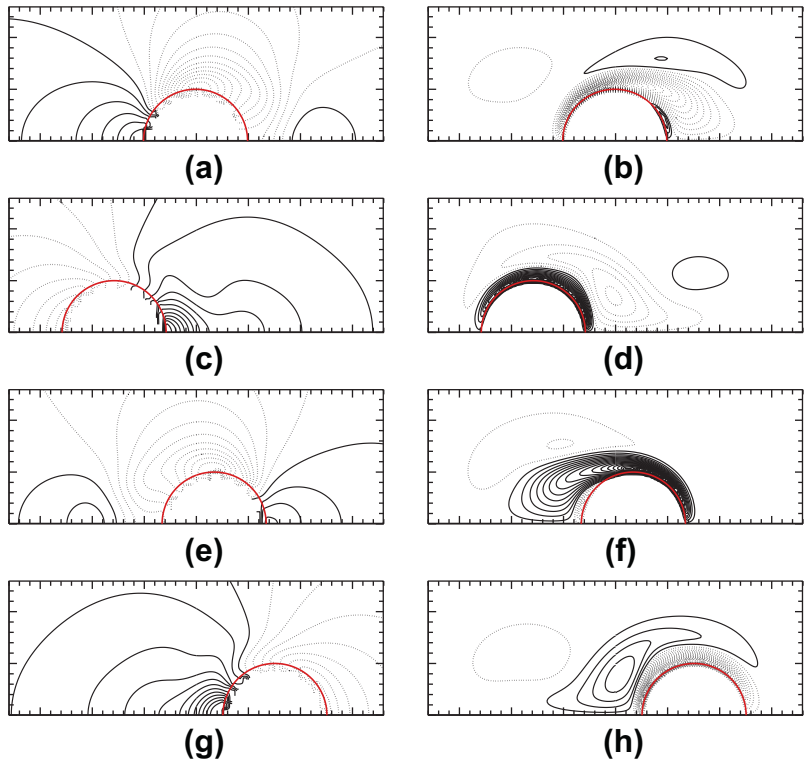

Fig. 5. In-line oscillating cylinder in a fluid at rest (Re = 100 and KC = 5). Pressure (right) and vorticity (left) contours at four different phase-angles. -1.1 < P < 1.1 with intervals of 0.09 and -26 < ω < 26 with intervals of 0.95. (a) 0°; (b) 96°; (c) 192°; and (d) 288°. Dashed lines correspond to negative iso-values.

angle $\theta = 90°$, the initial condition with symmetric boundary is a quarter of a circle with radius slightly larger than eight computational cells.

In Fig. 6, the shape of the drop is compared with the exact solution, drawn as a dashed line. It appears that the final shape of the drop depends in part on the location of the solid wall. Two cases are considered: in Fig. 6(a), the $\psi = 0$ line overlaps cell faces, whereas in Fig. 6(b) the line $\psi = 0$ passes through cell centroids. The wall-at-the-cell-face case corresponds to an almost flawless initial reconstruction of the half circle, which is essentially preserved in the rest of the simulation. The maximum variation (in absolute value) of the drop volume with respect to the initial condition is very small: 0.02%.

Conversely, the wall-at-the-cell-centroid case is the least ideal situation: the wall boundary treatment is unable to correctly reconstruct the curved surface, which appears flattened, at the contact point. This unbalance triggers a small oscillation ($\sim\Delta x/2$) of the drop shape, which eventually stabilizes in a slightly oblong shape. The maximum variation of the drop volume with respect to the initial condition is larger than in the previous case, but still rather modest for this grid resolution: 0.85%. The drop shape outcome is between the two described above when the solid wall traverses an intermediate position between the centroid and the cell face.

3.4. Effect of changing contact angle

A second set of tests consists in starting from the quarter of circle shape and assigning a contact angle different than 90°. The response of the drop is an indication of the algorithm behavior with a dynamic contact angle. The dependence on the position of the solid wall is examined for the 120° contact angle (Fig. 7) and for the 60° angle (Fig. 8). The notation and the initial conditions are the same as in the previous test. The initial drop radius is $R_0 \cong 8\Delta\xi$.

The steady-state solution is a truncated circle with the same initial area. The final radius, R_θ, is a function of the contact angle (in radians) $0 < \theta < \pi$,

$$R_\theta = \sqrt{\frac{\pi}{2\theta - \sin 2\theta}} R_0. \tag{19}$$

Clearly, $R_{\theta=\pi/2} = R_0$. The value of R_θ is used to draw the circular arc of the exact solution in Figs. 7 and 8.

The steady-state solution is reached within a few iterations. In the case of wall at the cell face, the average area variation with respect to the initial value is 0.13%, with maximum variation of 0.28%. For the same contact angle, but with the wall located at the cell centroid, the average variation is 4.1%, with the maximum variation of 7.7%.

A closer examination of Fig. 7(a) shows that the 120° angle is not represented exactly. The correct drop shape begins to be recovered in the first cell away from the solid wall, leaving an artifact of size Δx. The size of the artifact increases to 3/2 Δx in Fig. 7(b) because the location of the solid wall is just such that the cell containing the contact point is not active ($\psi_{i,j} < 0$).

For the 60° contact angle, Fig. 8 underscores that the cell containing the contact point does not resolve the correct contact angle, but that the drop profile away from that cell is captured correctly. As in the static case, there is a dependency of the result on the location of the solid wall. Particularly, when the wall is located at the cell face, the average value of area variation with respect to the initial value is 0.11%, with maximum 0.40%. With the wall located at the cell centroid, this variation is larger and similar to the value for the 120° contact angle case. The case in Fig. 8(b) is arguably the worst in terms of position of the solid wall: by displacing the wall downward by a tiny fraction of the grid size, the cell intersected by the solid wall becomes an active cell, and the simulation (not shown here) recovers area variations as small as the ones in the cell-aligned case.

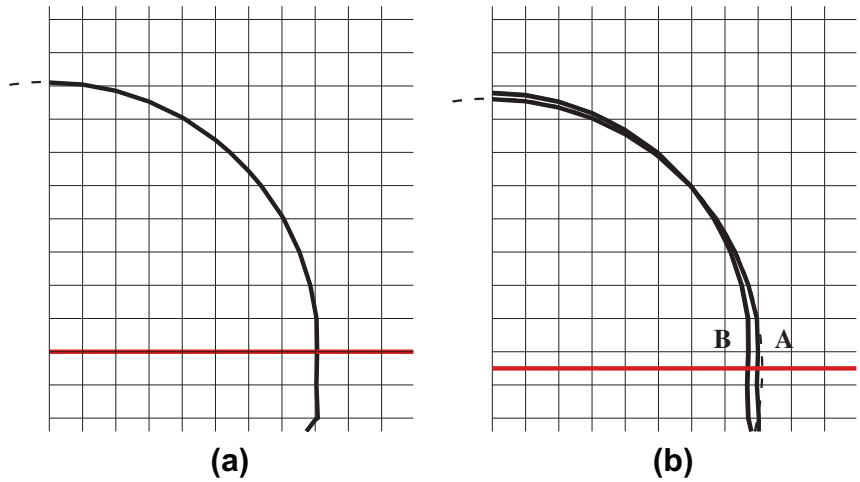

Fig. 6. Effect of wall location for static drop. The curved continuous lines are the $\phi = 0$ iso-contours at different times. In frame (a), the wall (thick horizontal line) overlaps the cell faces; in frame (b), the wall passes through the cell centroids. The exact solution is shown as a dashed circle line. The starting and prescribed angles are both the 90° angle.

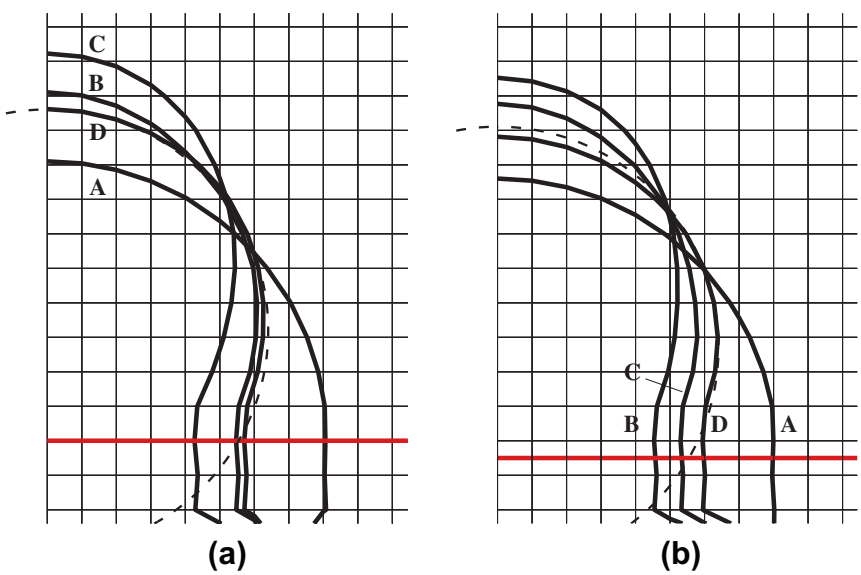

Fig. 7. Effect of wall location for a drop with 120° contact angle starting at 90°. The curved continuous lines are the $\phi = 0$ iso-contours at different times. In frame (a), the wall (thick horizontal line) overlaps the cell faces; in frame (b), the wall passes through the cell centroids. The exact solution is shown as a dashed circle line.

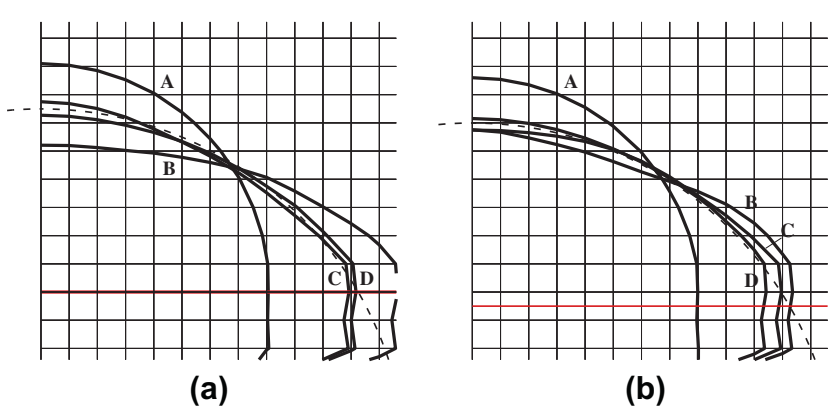

Fig. 8. Effect of wall location for a drop with 60° contact angle starting at 90°. The curved continuous lines are the $\phi = 0$ iso-contours at different times. In frame (a), the wall (thick horizontal line) overlaps the cell faces; in frame (b), the wall passes through the cell centroids. The exact solution is shown as a dashed circle line.

3.5. Convergence study for a two-dimensional static drop on an inclined plane

A more generic situation is when the wall is not aligned with a Cartesian direction. Since the previous tests indicated artifacts on a size at most comparable with the grid size, we expect to recover linear convergence with grid refinement even in this case. Contact angles of 90°, 120°, and 60° are considered again, but this time the solid wall is inclined at an angle of 18°. Starting from 8 cells across the drop diameter, grid resolution is increased by a factor of two until there are approximately 256 cells across the diameter, as shown on Table 3.

The metrics for this test are the percentage variation with respect to the exact drop height, e_0 (calculated in the normal direction to the wall),

$$e_0 = (1 + cos(\pi - \theta))R_\theta, \tag{20}$$

and the percentage variation with respect to the base length, L_0,

$$L_0 = 2sin(\pi - \theta)R_\theta. \tag{21}$$

To evaluate the simulation values of the height and base of the drop, the $\phi = 0$ iso-contour needs to be intersected with $\psi = 0$. This is accomplished by looking at the edges of the zero level of the function $\xi = \min(\phi, \psi)$ as a post-processing means to exclude the ghost region portion of the drop. The variations in e and L capture small shape deformations, so that a narrower base, for instance, corresponds to a larger height of the deformed drop. When there are 16 or more cells per diameter, area fluctuations are sufficiently small to be ignored in the analysis.

For $\theta = 90°$ the convergence of L to $2R_0$ is more than linear and the difference with respect to the exact value is quite small. A similar trend is found for the convergence of e to R_0. For instance, for $\Delta x/D = 1/32$, the drop recovers almost exactly the shape of a half circle, slightly narrower at its base (−1.3%) and taller (+0.16%) The small error and super-linear convergence rate are of course due to the 90° contact angle, which forces a condition close to symmetric boundary and therefore benefits from a curvature evaluation almost as accurate as that away from the solid boundary.

For contact angles larger or smaller than 90°, the discrepancy in height and base length increases substantially at the same grid resolution. For instance, if $\theta = 60°$ and there are only eight computational cells across the drop diameter, the drop shape is sufficiently deformed to suffer rather large deviations in L and e. However, with $\Delta x/D = 1/32$, the error becomes of the order of 1% or less. A similar behavior is observed for $\theta = 120°$. For both angles, the rate of convergence is linear, or less than linear but never below 0.5. Tests carried out with different plane inclinations confirm the reduction of the convergence order if the contact angle is not ninety degree. A similar outcome is reported by Afkhami and Bussmann (2009) for the contact line curvature of a three-dimensional drop resting on a Cartesian boundary: in that special setting, the curvature converged quadratically for $\theta = 90°$ and linearly for other contact angles.

3.6. Convergence study for a three-dimensional static drop on an inclined plane

The same study is repeated for a three-dimensional drop. The drop deformation from the correct truncated sphere solution (at contact angles of 90°, 120°, and 60°) is characterized by the height parameter e as well as by the length, L_a, and width, L_b, of the drop base. Results for the inclination angle of 18° are given on Table 4. At lower grid resolution the drop deformation is larger than in the corresponding two-dimensional case; errors decrease below 1% when the diameter is of the order of 64 grid cells for 90° contact, and at $D \sim 128\Delta x$ for the other two angles. Contrary to the results presented by Afkhami and Bussmann (2009), the maximum magnitude of spurious currents does not decrease with grid resolution, although their mean value over the computational domain does. However, the convergence of the parameters e, L_a and L_b to the solution is still quadratic for $\theta = 90°$ and linear for $\theta = 120°$ and 60°.

This test concludes a rather extensive examination of the proposed embedded boundary algorithm. The rate of convergence is on average first-order, and, when present, artifacts are only of the size of one grid spacing near the contact point. Thus, while simpler with respect to algorithms for contact point treatment with specially aligned boundaries, or for single-phase flow, the current staircase approach constitute a good starting point for further improvements. The two examples that follow demonstrate the robustness of the method in two-phase simulations of engineering flows.

4. Validation with Diesel injector data

The injector chosen for the validation calculation is the scaled-up Bosch six-hole diesel nozzle by Arcoumanis et al. (1998). The nozzle was manufactured at a 20:1 scale to allow flow measurements inside the sac volume and the injection holes; the nominal hole size of the model was 3.52 mm, corresponding to about 0.176 mm in the real injector. In the configuration used for the simulation, the conical element (the needle) was placed 6 mm above its seat, at the maximum lift of the second stage of the injector. A cross-sectional view of the sac and the cylindrical orifice tube is shown in Fig. 9. LDV measurements were taken in vertical planes across the injection hole; the planes were located at 9.5, 10.5, 13.5 and 16.5 mm from the axis of symmetry of the injector.

The test fluid in the experiment had density and kinematic viscosity of 893 kg/m^3 and $1.64 \cdot 10^{-6}$ m^2/s, respectively. The measured flow rate of the 6 mm lift configuration was $5.8 \cdot 10^{-4}$ m^3/s,

Table 3
Convergence for a two-dimensional drop on a plane inclined by 18° for three contact angles 90°, 120° and 60°; e is the height of the drop at rest and L is the base length.

	$\Delta x/D$=						Convergence rate
	1/8	1/16	1/32	1/64	1/128	1/256	
90°							
ΔL (%)	0.08995	−1.804	−1.296	−0.02686	−0.1076	0.02589	1.14
Δe (%)	2.097	−0.05861	0.1599	0.08282	−0.02847	−7.879e−4	2.69
120°							
ΔL (%)	8.6273	0.4063	0.5106	−0.8001	0.7584	−0.2524	0.632
Δe (%)	0.2148	0.8081	0.3105	0.5934	0.07909	0.06727	0.500
60°							
ΔL (%)	−18.99	−5.258	0.01349	1.098	−0.9131	−0.6358	0.735
Δe (%)	−1.971	−1.863	−1.045	−0.5870	−0.1089	0.03595	1.20

Table 4
Convergence for a three-dimensional drop on a plane inclined by 18° for three contact angles 90°, 120° and 60°; e is the height of the drop; L_a is the length of the drop base in the direction of the plane inclination; L_b is the width of the drop base in the perpendicular direction.

	$\Delta x/D$=					Convergence rate
	1/8	1/16	1/32	1/64	1/128	
90°						
ΔL_a (%)	−15.02	−5.489	−13.03	0.4954	0.2683	1.51
ΔL_b (%)	−14.86	−9.436	−2.233	−0.5102	0.07350	1.95
Δe (%)	−3.497	7.080	3.144	0.2674	3.680e−3	2.45
120°						
ΔL_a (%)	7.870	5.981	2.897	−0.1964	0.07159	1.18
ΔL_b (%)	7.149	9.873	4.162	2.589	0.3920	1.03
Δe (%)	1.793	−2.782	0.3301	0.2265	0.03560	1.49
60°						
ΔL_a (%)	−18.95	−27.08	−19.70	−3.754	−1.548	1.01
ΔL_b (%)	−18.80	−3.876	8.253	2.215	−0.5921	1.08
Δe (%)	3.926	5.230	2.777	−1.146	0.09300	1.30

corresponding to the average velocity of 10.2 m/s inside the orifice. Based on this velocity and on the orifice diameter, the Reynolds number of the experiment was approximately $2 \cdot 10^4$. The surface tension coefficient in the simulation was 0.024 N/m; the static contact angle was 90°.

The pressure difference across the transparent injector was 0.85 bars, with a total discharge coefficient of the needle seat and injection holes of 0.78. Because the pressure in the recirculation zone at the entrance to the hole could decrease below the vapor pressure of the liquid, the onset of cavitation was found to be possible and to strongly depend on the needle lift and eccentricity.

Cavitation is not modeled in the simulation presented here. Another important difference with respect to the experimental setup is the absence of the expansion tube to collect the liquid after passing the orifice: in the simulation, the liquid is injected into air at standard conditions. This setting may affect the exit velocity of the jet, but has the advantage of showing the direct link between spray characteristics and nozzle internal flow.

For ease of discussion, we will refer to the side where the sac is located as "bottom", in the negative y direction of Fig. 9. To reduce the computational cost, the simulation includes only a quarter of the injector and one of the six orifices. This configuration is displayed in Fig. 10. The domain of $100 \times 100 \times 40$ mm is sufficiently large to include the development of the jet for several orifice diameters downstream of the orifice exit. With a coarse mesh of $80 \times 80 \times 32$, the four levels of adaptive refinement used in the simulation provide a finest grid resolution of 0.0781 mm. A uniform inlet boundary velocity is applied along the open area at the top of the injector to deliver a sixth of the mass flow rate reported in the experiment. Here and in the next example, no turbulence model is used.

The injector walls are described by a triangular mesh of 28,800 elements, with initial smallest grid size of 0.4 mm. Within one solution cycle, the handling of the solid wall requires approximately 5.7% of the CPU time, much less than the treatment of the gas–liquid level set (34.7%), the pressure solve iterations (33.4%), and the solution of the nonlinear advection term (15.5%). This cost does not include the calculation of the solid level set, since it is performed only once on the auxiliary grid at the start of the simulation. In the interval used for this analysis, the average total of all the computational cells was $3.7 \cdot 10^7$. Calculations were carried out on the Redsky Sandia cluster using (on average) 24 SUN X6275 blades (2.93 GHz dual socket/quad core configuration with 12 GB RAM per compute node) for a total of 192 cores.

From the simulation, we find that a time-averaged pressure drop of 0.75 bar takes place through the portion of the injector that is included in the calculation. This value compares well with the value 0.78 reported in the experiment. The highest flow velocities in the internal flow occur at the upper corner of the hole inlet, where the lowest fluid pressures can be expected. This can be seen in the snapshot of the vector field shown on Fig. 9. In the picture, only one in twenty vectors is shown, for clarity; the continuum black line is the intersection of the $\psi = 0$ iso-contour with the plane slicing the domain. In the snapshot from the CLSVOF simulation, only one in twenty vectors is shown, due to the much greater grid density than in the RANS calculation; the continuum black line is the intersection of the $\psi = 0$ iso-contour with the plane slicing the domain.

Examination of several simulation snapshots suggests that the recirculation zone re-attaches approximately past half of the hole. For a more quantitative analysis of the flow, in Fig. 11 the mean and root mean square (rms) values of axial velocity are compared to the LDV measurements. The plots are drawn along the orifice diameter from the four cross-sections shown on Fig. 9 and normalized by the mean orifice velocity of 10.2 m/s. In each plot, the distance of the axial coordinate is counted from the top of the cross-section and normalized by the diameter of the orifice.

The velocity mean values, in the left column of Fig. 11, compare favorably with the experiment; however, the measured recirculation zone is broader and longer than in the simulation: faster acceleration of the computed flow can be seen at $x = 13.5$ mm. Where data points exist for comparison near the orifice wall, the computed boundary layer appears thicker: this aspect can be attributed to insufficient grid resolution there. Note that the average wall shear velocity, $u_\tau = (\tau_w/\rho)^{1/2}$ for this Re = 10,000 case is of the order of 0.02 m/s, corresponding to a location of the first point away from the wall of $r^+ = \Delta x\, u_\tau/\nu \sim 1$ in wall units.

Pressure fluctuations with respect to the average cross-section value (not shown here) are mostly positive in the upper half of the exit orifice, consistently with the lower mean axial velocity in that region. Occasionally, negative pressure variations are also observed in the upper half because of flow unsteadiness. The LDV rms values of velocity are consistently larger than in the calculation, with the largest discrepancy in the recirculation zone (this rms peak is similarly lost in the RANS calculation from Arcoumanis et al., 1998). Velocity fluctuations near the orifice's walls are actually of the correct magnitude, but they decrease toward the center of the orifice. This defect is common to under-resolved simulations that attempt to capture the unsteadiness of turbulent flow without a sub-grid model. It is also possible that cavitation near the entrance of the orifice contributes to the large fluctuations there. A cavitation model will need to be added in future validation studies.

Considering now the coupled process of spray formation, in Fig. 10 a few drops larger than the orifice diameter can be observed

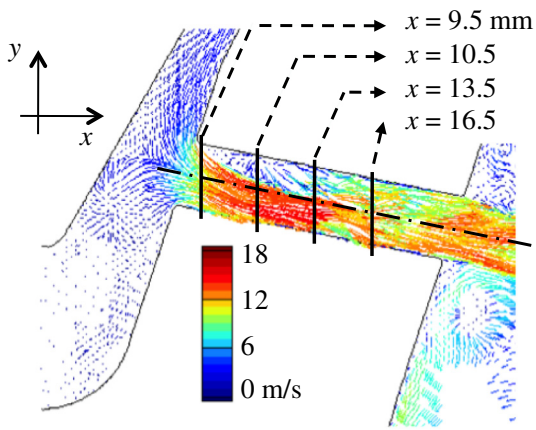

Fig. 9. Vector flow field on a domain cross-section (instantaneous snapshot).

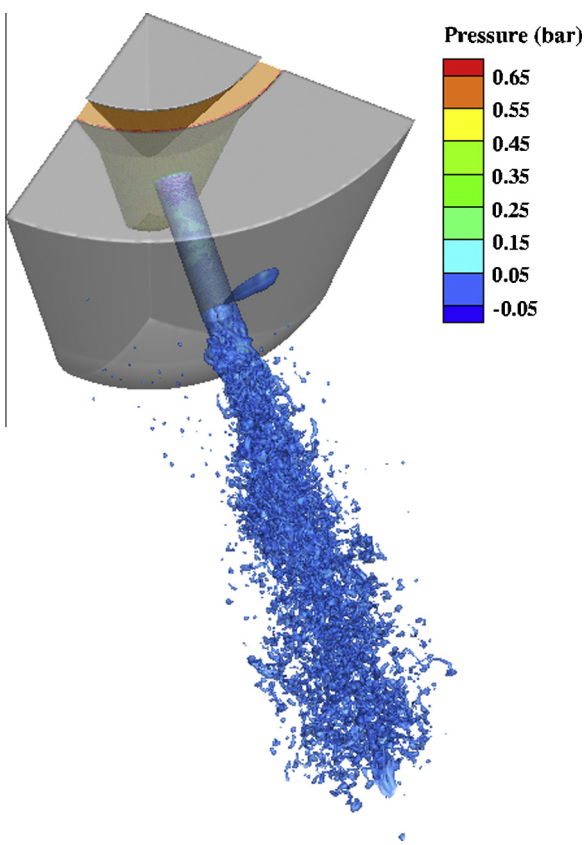

Fig. 10. Terminal tip of the injector. Simulation snapshot at time $t = 35$ ms of the full domain.

in the initial build-up of the jet, with one large drop attached the injector's wall. Droplets begin to detach from the jet approximately two orifice diameters downstream of the hole, and from there a complex network of ligaments and drops rapidly consumes the liquid core.

The most apparent feature of the jet outside the nozzle is that it slightly bends upward (in the positive y direction in Fig. 9), away from the tip of the injector. This is due to the occurrence of larger velocities in the bottom half of the orifice compared to the upper half. As the faster side of the jet spreads less outwardly, it is relevant to interrogate the simulation about the corresponding drop size. Droplets are identified as isolated blobs from the $\phi = 0$ liquid surface. Their volume is translated into the effective diameter of the same-volume sphere, producing a histogram of diameter distribution. The one shown in Fig. 12, for a snapshot taken at $t = 35$ ms, seems to suggest that the half of the spray that is closest to the tip of the injector has fewer droplets in the range from 200 to 400 μm. At this point, as with the other results reported in this section, more data would need to be collected before quantitative statements can be made.

5. Demonstration with moving parts

The last test consists of an instance of rigidly moving wall that is used to demonstrate the flexibility of the new method. The test takes advantage of current efforts by the Engine Combustion Network (ECN) to characterize the internal geometry of automotive injectors. Five nominally identical Diesel Bosch injectors with single axial orifice form the baseline that is presently identified as "Spray A" (ECN, 2013). X-ray tomography, X-ray phase-contrast

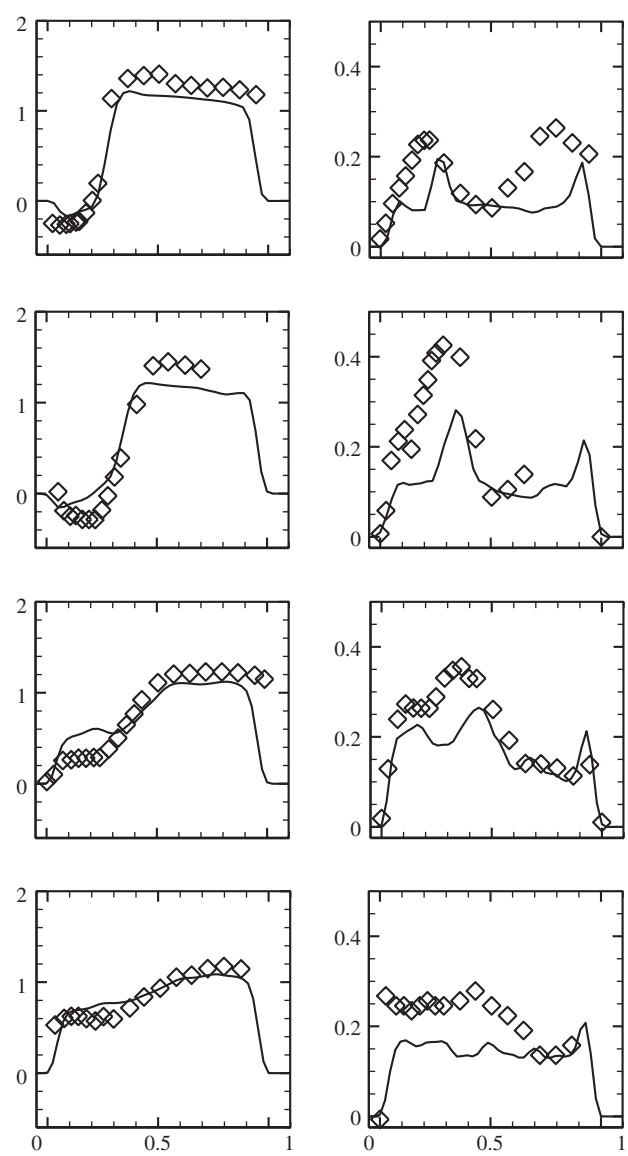

Fig. 11. Comparison with LDV measurements along the injection hole. Left column: comparison of the mean values of axial velocity at four cross-sections along the injection hole (symbols are measured values, continuous lines are from simulation). Right column: comparison of the root mean square values of axial velocity. From top to bottom: $x = 9.5$, 10.5, 13.5, and 16.5 (see Fig. 9 for reference).

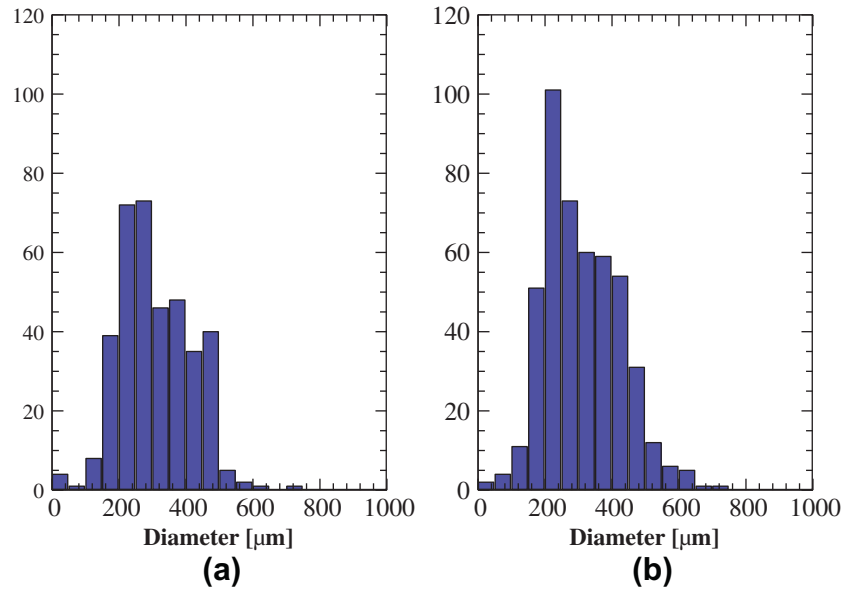

Fig. 12. Droplet size distribution in the half the spray that is closer to the injector's tip (a) and farther away (b) at $t = 35$ ms.

imaging, silicone molding, and optical microscopy are combined in this effort to define a database of the internal geometry (including deviations from the nominal shape) in a format that is immediately accessible to simulation. The ECN makes also available a trajectory file with the three-dimensional motion of the needle determined from X-ray phase-contrast as recorded by a high-speed camera.

Three snapshots of the needle motion are displayed in Fig. 13. The needle tip (in green[1]) translates according to the specified trajectory with respect to the injector cap. For instance, in its fully open position in frame (a) the base of the needle is outside of the computational domain, to the left of the dashed line: the lift of the needle from its seat opens an annular passage to the liquid. Conversely, in frame (b) the needle base is almost completely inside the computational domain. Frame (c) shows the deviation of the needle motion from a perfectly axial trajectory and the resulting non-axisymmetric annular passage. Interestingly, the SN 210675 Bosch model used in this test is affected by slight manufacturing defects (the reader is referred to the ECN web site for full documentation). As in all the other injectors of the Spray A batch, off-center position with respect to the sac, tapering of the hole, and deviation of its cross-section from a perfectly circular shape of diameter 90 μm can be observed. These geometrical characteristics are discussed by Kastengren et al. (forthcoming).

The needle velocity, necessary to populate the solid ghost cells, is calculated by differentiation of the displacement values in the trajectory file. The measured injection pressure of 155 MPa is applied directly as constant boundary value at the inlet boundary of the injector. A constant exit pressure is applied to the other five sides of the computational domain box. The gas phase has density of 22.8 kg/m^3, corresponding to an ambient pressure of 6 MPa at 900 K; the liquid gas/density ratio is therefore 30.8. The dynamic viscosity of the liquid is 0.00613 g/cm-s; the dynamic viscosity of the gas is 0.00022 g/cm-s. The surface tension coefficient used in the simulation is 0.024 N/m. The static contact angle is 90°.

The base computational domain is a $64 \times 64 \times 192$ box with the longest side oriented along the injector's axis measuring 5.1 mm. Two levels of refinement are added to obtain a minimum grid spacing of 6.64 μm. The resulting 13.5 cells per orifice diameter, while adequate for this simple demonstration, are obviously not sufficient for an in-depth study of Diesel fuel injection dynamics during this transient.

CPU timing during the simulation shows that, within one solution cycle, the handling of the solid wall requires 47.3% of the CPU time and it is now comparable to the pressure solve iteration (45.4%). The third most expensive item is the solution of the nonlinear advection term (4.4%). The average sum of all the computational cells is $1.0 \cdot 10^7$. The increased CPU time percentage is due to the calculation at every time step of the needle level set ψ_2, which is carried out on an auxiliary grid of $93 \times 93 \times 144$. The full reconstruction of ψ_2 was not strictly necessary in this case because of the simple translating motion of the solid body: a faster algorithm for this task is under implementation.

The representation of the injector's moving geometry and the pressure boundary condition enable a realistic calculation of the mass flux variation in time through the orifice. To illustrate this point, Fig. 14 shows the pressure field on a section plane passing through the axis of the injector in a close-up of the full computational domain. The black line is the intersection of the plane with the instantaneous jet surface; the red line is the intersection with the solid wall. Contours of pressure inside the solid domain are due to the overlapping of refinement levels (therefore their variation in time bears no physical significance).

The first frame in Fig. 14 captures a moment before the needle returns to its seat, when pressure in the liquid phase gradually decreases between inlet and orifice exit. Because the simulation reflects the side displacement of the needle observed in the experiment, the closing of the needle is asymmetrical., The second frame displays the partial closure of the flow in the upper half (negative y direction in the plot reference) of the injector. The flow upstream of the contact line finds its way only through the bottom passage, and the full pressure drop takes place along that constriction. The jet outside of the orifice at this point still maintains a turbulent appearance, with drops forming a few diameters downstream.

In the third frame of Fig. 14 the passage between needle and cap has closed and the mass flux through the pressure inlet has dropped to zero. The jet carries a residual momentum, but its appearance has reverted to laminar and drop formation is much delayed. A large chunk of liquid is at the point of detaching from

[1] For interpretation of color in Fig. 13, the reader is referred to the web version of this article.

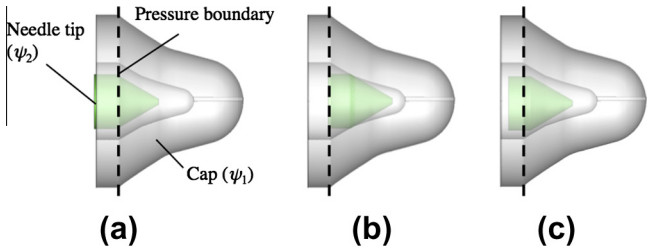

Fig. 13. Relative motion of the two injector's components. The dashed line is the trace of the boundary plane in the simulation.

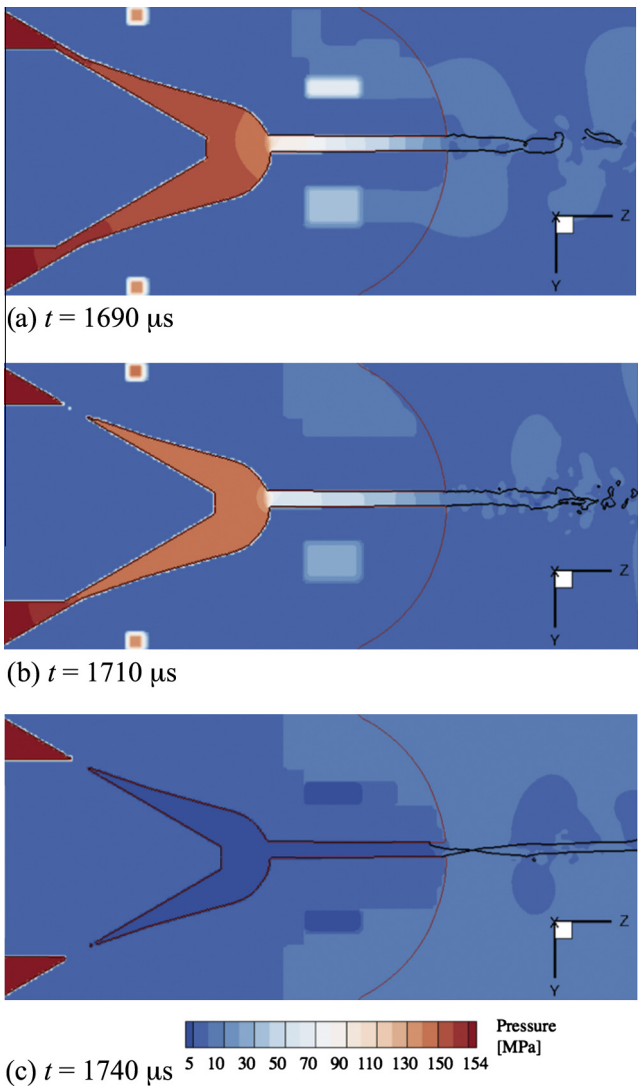

Fig. 14. Snapshots of plane sections through the injector axis. The black line is the intersection with the jet surface; the red line is the intersection with the solid wall. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the remaining liquid in the sac. Note that the pressure in the sac and in the orifice is slightly lower than in the outside environment: droplets were actually observed in the experiment as they were sucked back in the orifice. The flow at these conditions is very rich in features and deserves further analysis in a following work.

6. Summary and path forward

A new simulation capability is presented where internal and external flows can be calculated seamlessly across an injection orifice. The external gas–liquid interface is handled by the combined level-set volume-of-fluid (CLSVOF) method and the solid wall boundary is represented by a different level-set function on the same block-structured grid. This approach shifts the burden of handling complex internal geometries to the much simpler task of obtaining a tessellation of the injector surfaces, therefore side-stepping volume mesh generation altogether.

The numerical technique for handling solid boundaries follows in the group of Cartesian grid methods with embedded boundaries, but the capability of modeling at the same time the formation of the liquid jet, including primary atomization and air entrainment effects, is unique. The embedded boundary treatment appears to be at best first-order accurate, but improvements could be borrowed from single-phase methods.

Moving parts of the injector and contacts, such as they occur when the needle shifts during operation cycle, are trivial to include because they simply require a level-set recalculation. The update of the ghost region cells and the handling of the contact angles are relatively inexpensive operations, as shown by the example discussed in Section 4. There is a substantial extra cost when the full reconstruction of the solid level set is carried out at every time step, as in the moving needle case discussed in Section 5. A technique such that it is only deformation, and not body translation or rotation, to require full reconstruction of the solid level set is under development.

On the other hand, the embedded approach is extremely convenient from a performance point of view because it preserves the symmetric form of the coefficient matrix of the variable coefficient Poisson equation. A fast Poisson solver, such as the one we use in our simulations, is crucial in multiphase flow simulations with large density ratios.

Further work will address the physics associated with internal flow in real-size injectors, such as cavitation and liquid phase compressibility. Reaching sufficient grid resolution to provide a realistic turbulent flow at high Reynolds numbers remains an outstanding challenge, together with the fact that high-resolution liquid injection and spray formation remain "stiff" simulations, whose stable timestep is several orders of magnitude smaller than a full injection period.

We conclude by noting that, with the increased accuracy enabled by injector modeling, the full description of the injector's geometry and operation is more than ever crucial for a correct comparison with laboratory measurements.

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